

ELITE SUMMER PROJECT

Analyzing the Impact of Eating Habits and Sedentary Lifestyle on Health Risk Patterns

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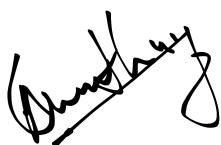
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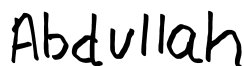
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This is to certify that Aditya Kumar and Abdullah Nasim, students of B.Sc.(Hons.) Computer Science IInd Year, and Shubhi Srivastava, student of B.Sc.(Prog.) Physical Science with Computer Science IInd Year have completed their research project titled “Analyzing the Impact of Eating Habits and Sedentary Lifestyle on Health Risk Patterns” to partially fulfill the credit for the ELITE Summer research program carried out at ACHARYA NARENDRA DEV College, University of Delhi, during Mid June-Mid August 2026. The project submitted is a record of original research work carried out by Aditya Kumar Chaudhary, Abdullah Nasim and Shubhi Srivastava and is not submitted elsewhere.



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ABSTRACT

Today, **sedentary lifestyle** is a very common habit among young people and adults. Lifestyle and health are affected by many factors such as *daily habits, physical activities, stress, sleep well-being, and other lifestyle habits*. If we observe these factors together, we can identify some patterns that may not be visible when these factors are studied separately. Therefore, this study examined the lifestyle habits using two datasets. The primary objective of the study was to identify correlations between diseases and lifestyle habits, analyze patterns associated with disease occurrence in relation to lifestyle habits, and suggest steps that may help individuals adopt small changes to maintain a healthy lifestyle.

The aim of this project was to investigate the relationships between lifestyle and health-related variables. Thus, both data sets were used for a wider perspective and allow for a more complete analysis. Overall, we observed that **lifestyle habits** can contribute in identification of major health risks at an early stage. Also these datasets suggest that one single factor cannot explain health risk. Thus, the combined analysis of the features provide a useful base for understanding of these patterns and further analysis of health outcomes.

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Chapter 1

Introduction

In this chapter the core problem statement and the core motivation is discussed which made authors think about how simple lifestyle habits can help grouping people together, along with listing the core objectives and aim of the research.

1.1 Motivation

Health is influenced by various combinations of biological, behavioral, and lifestyle-related factors. In recent years, changes in eating habits and daily physical activity have become an important area of concern because unhealthy diet patterns and sedentary lifestyle might contribute to the development of various health-related conditions. Factors such as *calories intake, sleep pattern, physical activity, daily stress, social relationship, body mass index (BMI), blood pressure, heart rate, daily workout, meditation, and other lifestyle characteristics can interact with other and may* collectively indicate different levels of health risk [1]. Understanding these relationships is therefore important for identifying patterns that may not be apparent when individual factors are considered separately.

The increasing availability of health-related datasets provides an opportunity to study these relationships using data exploration and machine learning techniques. Rather than examining a single factor in isolation, a data-driven approach can be used to investigate multiple health and lifestyle variables together, to identify groups of individuals with similar characteristics. Such analysis provides a broader understanding of how **behavioral** and **physiological factors** are associated with health risk patterns and can help predict these health risks so that appropriate medications or physical activities can be adopted to cure these health conditions early [2].

Traditional analysis of health data often focuses on individual variables or predefined health outcomes. However, the approach overlooks relationships among multiple variables or their interaction. A multidimensional data analysis approach can overcome this limitation by considering dietary, physical activity, physiological, and other lifestyle related variables together [2].

1.2 Problem Statement

The relationship between lifestyle behavior and health risk is multidimensional and can not be understood adequately by examining individual health indicators separately [3]. Eating habits, physical activity, sedentary behavior, sleep, body composition, cardiovascular measurements, mental wellness, lifestyle problems and other hidden factors may interact differently and produce different **health profiles** across the population among different individuals. The problem addressed in this study is to determine the feasibility of identifying meaningful and distinguishable health-risk patterns from combinations of lifestyle and health related variables. Also, this study seeks to examine whether grouping of individuals according to similarities in their health characteristics might reveal meaningful differences or associations with important health outcomes.

1.3 Objectives

This study is trying to provide a deep understanding of interconnection of sedentary lifestyle habits with their health. In order to achieve this we will :

- Conduct exploratory data analysis (Uni-variate, Bi-variate, Multi-variate and Outlier analysis) to understand the **distribution, relationships, and quality** of the underlying health and lifestyle features.
- **Engineer meaningful features** to better understand health profiles such as the mental wellness index, social wellness index, physical wellness index, composite lifestyle risk score, and recovery and wellness index.
- Apply dimensionality reduction (Principal Component Analysis) to the standardized health features in order to visualize the structures of underlying data.

- Identify natural health profiles within the population using unsupervised clustering techniques (K-Means and DBSCAN) to **reveal structural patterns and distinguishing features**.
- Draw evidence based conclusions regarding the relationship between **lifestyle patterns and chronic risk disease**.

1.4 Scope of the work

The scope of this study is focused on the analysis of health and lifestyle related datasets to identify patterns associated with eating habits, physical activity, sedentary behavior, and other relevant meaningful health indicators. The study incorporates **two datasets** analyzed by different researchers to examine common research questions from different data perspectives.

The proposed work analyzes the impact of sedentary lifestyle and day to day habits on health, and aims to identify patterns associated with the lifestyle habits. The proposed work identifies patterns and associations rather than to establish causal relationships. Therefore, the **findings should be interpreted as evidence of relationships and patterns** within the analyzed datasets instead of treating them as proof that a particular eating habit or lifestyle behavior directly causes a specific health condition. Differences in variables, sample characteristics, and data sources across the datasets are also taken into consideration while making conclusion on health risks related to lifestyle issues.

1.5 Organization of the report

This section describes the organization of the report in brief. Chapter 2 describes the **study of available research papers** related to the topic and the gap present in those studies. Chapter 3 **describes the methodology used** in the Data collection, Preprocessing, EDA, feature engineering, Feature Categorization, Principle Component Analysis and Clustering. Chapter 4 **lists down the result** of the experimentation performed along with the analysis of results. In the last chapter 5 the **conclusion of the whole study** is written along with explaining the future scope associated with the research.

Chapter 2

Related Works

This chapter discusses the related literature on health risk related to lifestyle, followed by research gap and relevance to the present study.

2.1 Literature Review

There is a growing acknowledgment that health risks related to lifestyle are the consequence of the interaction of behavioral, physiological and environmental factors, rather than the effect of a single variable. Dietary behavior, physical activity, sedentary behavior, body composition, sleep and cardiovascular indicators may be of particular importance to the development of obesity and other non communicable diseases. Therefore, **a multidimensional viewpoint** is the foundation for studying health risk patterns as combinations of lifestyle and physiological characteristics, instead of as isolated individual risk factors.

González-Gross and Meléndez [4] reviewed the relationship between sedentary behavior, active lifestyles, sport and obesity prevention with emphasis on the health benefits of regular physical activity. Their review links physical activity to energy expenditure, body composition, cardiovascular health and prevention of overweight and obesity. The evidence shows that health benefits can be obtained from both moderate and vigorous physical activity and that it is important to maintain an active lifestyle for the control of body weight and reduction of cardiovascular risk. The authors also emphasize the significance of the energy balance, which is a conceptual link between dietary intake, physical activity and changes in body composition [4]. These observations are directly relevant to the present study as variables such as calorie intake, BMI, daily steps and exercise related measures are considered

together when investigating health related patterns.

The connection between lack of physical activity and chronic diseases are also investigated. Park et. al. [1] highlight the difference between physical inactivity and sedentary behavior. According to authors, sitting longer is independently associated with several adverse health outcomes. The review demonstrated that sedentary conduct was linked to cardiovascular disease, type 2 diabetes, hypertension, dyslipidaemia, cancer, musculoskeletal disorder, and mortality. Moreover, the authors emphasize that the negative consequences of prolonged sitting are reflected in physically active people, which indicates that this type of behavior is unhealthy on its own [2]. The review highlights the importance of interruption in long sitting duration with light and moderate exercise, as such physical activity can reduce metabolic and cardiovascular risk. So, the variables associated with physical exercise and physiological function can contribute to multidimensional health risk analysis.

The broader context for health related risks is provided by the Segovia Consensus Document on Obesity and Sedentarism in the 21st Century [5]. The authors discuss obesity and sedentarism as multifactorial issues, arguing that the etiology of these conditions cannot be explained exclusively by caloric intake and lack of physical exercise. Moreover, the review highlights sleep, social and demographic factors, tobacco consumption, pharmacological agents, and overall lifestyle as important determinants of overweight, obesity, and physical inactivity [5]. In addition, the review underlines the complex interconnections between nutrition, physical exercise, body weight, and health status, as well as the need for combined interventions [5]. Particularly relevant to the current study, the consensus highlights the importance of analyzing all components of energy balance rather than focusing on individual elements, such as dietary habits and levels of physical activity, separately.

Paraschiv [6], who reviewed the impact of sedentary lifestyle on students' physical and mental health, also emphasized the role of sedentary behavior in health status. The key point discussed in this article is screen time allocation as a factor that promotes sitting and contributes to developing obesity and cardio-metabolic disorders. Urbanization, techno-anthropology, and the increasing number of occupations requiring sitting are also highlighted as determinants of sedentary behavior [6]. Physical inactivity is linked with higher risk of obesity, cardiovascular disease, type 2 diabetes, musculoskeletal issues, anxiety, depression, and poor sleep [6]. While students' lifestyles are not the primary focus of this paper, the review illustrates the multidimensionality of

the health impact of a sedentary lifestyle.

Finally, the multi-factorial nature of sedentarism was reported by Goyal and Rakhra [7], who updated the understanding of the health effects of a sedentary lifestyle. The review defines sedentary behavior as low energy expenditure activities, such as sitting and viewing the screen, and discusses the mechanisms of action, patterns, and factors of this type of activity. The focus of the review is on the associations between prolonged sitting, physical inactivity, and cardio-metabolic syndrome [7]. However, the paper also discusses several chronic diseases and conditions that may be promoted by a sedentary lifestyle, including obesity, type 2 diabetes, metabolic syndrome, cardiovascular disease, cancer, musculoskeletal disorders, and psychological conditions [7]. In addition, the review highlights the importance of considering the combined effects of physical activity and sitting since moderate exercise can reduce some cardio-metabolic risks, such as lipid profile deterioration, insulin sensitivity, adiposity, cardio respiratory fitness, and inflammatory activity, counteracting the negative effects of prolonged sitting [7].

Therefore, the literature has increasingly focused on developing a more integrated understanding of lifestyle-related health. Instead of treating diet, physical activity, sitting time, sleep, and physiological measures as separate outcomes, these variables can be thought of as part of a larger lifestyle domain. Such approaches are particularly useful in population-level studies where participants may display a variety of favorable and unfavorable patterns on different variables. For example, an individual may show high levels of physical activity but increased body mass index (BMI) and blood pressure, while another may be physically inactive but display healthier cardio metabolic readings. Therefore it is important to consider combinations and patterns of behavior when looking into lifestyle related health.

Recent advances in digital health and artificial intelligence (AI) offer another valuable approach to studying multidomain lifestyle health. Romero-Tapiador et al. [2] developed the AI4FoodDB database for personalized nutrition and lifestyle research, which includes a wide range of anthropometric, lifestyle, and health assessments, as well as nutritional, physiological, and digital measures. The paper describes the database’s organization and structure, which can serve as an inspiration for similar projects, and provides an example of a participant-level analysis to demonstrate the value of the individual measures and their combinations for health research [2].

The study therefore helps illustrate the potential of lifestyle research

with large datasets [2]. AI4FoodDB is particularly valuable in showing how lifestyle factors like diet, physical activity, and sleep can be combined with physiological and digital measures. The database contains variables like BMI, calories, alcohol consumption, tobacco consumption, physical activity, sleep, heart rate, blood pressure, step count, nutritional data and other variables for overweight and obese participants [2]. The authors emphasize the importance of synthesizing these different measurements obtained with different methodologies in the investigation of complex associations between lifestyle behaviors and health [2]. It then shows with data from one participant how dietary, physical activity, sleep, heart rate and glucose variables can be analyzed together rather than as separate domains.

2.2 Research Gap and Relevance to the Present Study

Together, the reviewed literature provides strong evidence that dietary behavior, physical activity, sedentary behavior, body composition, sleep and physiological characteristics are dimensions of health that are interconnected [1, 2, 4–7]. However, much of the current literature has been directed toward the relationship of sedentary behavior or physical activity with specific health outcomes, physiological mechanisms or intervention and prevention strategies. Other work such as AI4FoodDB are focused on building richer multi modal datasets and technological infrastructure for future personalized-health applications.

This provides us an opportunity to adopt a data driven approach for pattern recognition. This study is not about one risk factor or one outcome, but rather about many lifestyle and physiological variables at once. The goal is to see whether people can be grouped into meaningful categories based on similarities in their health and lifestyle profiles. Therefore, the proposed study uses exploratory analysis, feature engineering, dimensionality reduction through Principal Component Analysis (PCA), and unsupervised clustering to investigate the underlying structure of health-related data. This approach is in line with the increasing literature on the multi-factorial nature of obesity, sedentary behavior and lifestyle-related disease, but extends it toward the identification of multivariate health-risk profiles.

Crucially the present study attempts to detect patterns and associations, not to establish causal relationships. The literature does support the existence of relationships between lifestyle characteristics and health outcomes,

but the observational and cross-sectional nature of the datasets used here preclude the interpretation of the patterns identified here as evidence that a particular lifestyle factor directly causes a disease outcome.

Chapter 3

Methodology

This chapter explains the exact process used for data cleaning and exploratory data analysis to capture hidden patterns in data.

3.1 Data Collection & Specifications

Two datasets are used in the current work for identifying health risks associated with sedentary lifestyles.... Dataset 1 contain 15972 observations and 24 variables [8]. This dataset focuses on physical activity, dietary behavior, stress, social relationship, productivity, recovery, personal well being, and work life balance. This dataset was formed using the questionnaires and contained rating scales(0-2, 0-5, 0-10) of different lifestyle habits.

The second dataset, a synthetic data, originally contained 1,000 observations and 16 variables [9]. The dataset covers demographic characteristics, body measurements, physical activity, dietary and lifestyle behavior, physiological indicators, and health outcomes for individual.

3.2 Dataset Preprocessing and Cleaning

It is well known fact that preprocessing is required for data cleaning so that factual information could be captured using EDA and Machine Learning (ML) approach. The following steps are used for data cleaning [10].

- i. **Handling Missing Values:** Missing or null values can make the analysis inaccurate, and it should be properly dealt with before making the analysis using machine learning algorithms. Most of the ML algorithms cannot accept NaN or null values as input. The dataset was

cleaned by eliminating missing values so that it would be compatible with the algorithms used in this study.

- ii. **Removing Duplicates:** Repeated duplicate records in the dataset may create biasness for a specific group. The ML model will assign more weightage to repeated records, which may not be revealing useful information and contributing much in model building.
- iii. **Handling Outliers:** To prevent extreme values from distorting model training, skewing mathematical calculations, and causing poor predictive performance on normal data, outliers are generally removed from the dataset.
- iv. **Encoding Categorical Values:** Machine learning model strictly operates on a pattern of data values for any specific output or any specific task. When we will feed string to those models then they will not give the desired output. So categorical values needs to be encoded into number to be used in those ML algorithms.

3.3 Exploratory Data Analysis

EDA is used to understand the feature distribution, relationship between variables using correlation, trends and patterns identification using feature engineering and clustering. Uni-variate, Bi-variate, Multivariate and Correlation analysis are useful to understand the underlying structures in the data.

3.4 Feature Engineering

It is used to create new meaningful and interpretable features from the original variables for better representation of the given lifestyle dimensions so that important information could be revealed using clustering. Engineered features are clubbed to represent a collective health indicator that was originally missing in original data. Following three steps were used for the feature engineering.

- i. **Feature Selection :** All the features in Dataset 1 [8] were on different scales like some of them were on rating of 0-2, 0-5 and 0-10 which creates non uniform distances between them which will create problem while using the clustering method using the euclidean distance and in

order to get rid of this problem all the features were normalized on a scale of 0-1 using the theoretical scaling method.

- ii. **Feature Creation :** After feature selection we created new features using selected features. In this step a total of 6 features were created and a total of 12 features (6 original features and 6 engineered features) for validation. And a total of 4 features were created in Dataset 2. [9]
- iii. **Feature Validation :** The six newly created features (*Productivity Index, Mental Wellness Index, Recovery & Wellbeing Index, Social Wellness Index, Stress Burden Index and Physical Wellness Index*) and 6 original feature was selected for validation. In this phase the features were checked whether they contain redundancy, they are from valid range, are the statistically correct and should they be kept, modified or removed.

3.5 Feature Categorization

The features work categorized into two categories: original and engineer features; here we have categorized features because it will benefit in clustering and we will be able to visualize these features in a better way. After categorization the features were preprocessed to be used for PCA and clustering. The data was first scaled using standard scaling and then was checked whether it contained any error caused due to scaling.

3.6 Principal Component Analysis

Principal Component Analysis (PCA), an unsupervised machine learning technique, is used to compress a high number of features into lesser features, which helps to better visualize patterns and identify them. The objective is to have reduced components that capture 70 to 90% of the variance in the original data and are capable in revealing underlying structural patterns using clustering.

3.7 Clustering

Data points are clustered into specific groups for identifying similar features exhibited by data points which can further be analyzed to identify patterns among them. We use two clustering techniques viz. K-means and

DBSCAN for cluster identification. Afterwards, clusters are evaluated using Within-Cluster Sum of Squares to find the optimal clustering scheme to reveal number of clusters. Additional metrics like Calinski Harabasz score, Davies Bouldin score and Silhouette score are also used for scheme evaluation.

Chapter 4

Experimentation and Result Analysis

4.1 About Data

We have used two datasets named as Dataset 1 [8] and Dataset 2 [9] whose descriptions are given in Table 4.1.

Table 4.1: Description of datasets used

Parameter	Dataset 1 [8]	Dataset 2 [9]
Dataset	Lifestyle and Health Dataset	Health Activity Dataset
Total observations	15,489	800
Final clustering variables	12	7
Analysis type	Unsupervised / exploratory	Unsupervised / exploratory
Clustering algorithm	K-Means	K-Means/ DBSCAN
Final PCA components	3	3
Final number of clusters	2	4

4.2 Implementation Details

Following Python environments and libraries are used for exploratory data analysis and implementation:

- Python environment: i. Python 3 ii. Jupyter Notebook / Vs code
- Packages for Data manipulation: i. NumPy ii. Pandas

- Packages for Plotting: i. Matplotlib ii. Seaborn
- Data Analysis: i. Scikit-learn ii. SciPy iii. Statsmodels

Table 4.2 gives the details of hardware configuration used for proposed analysis.

Table 4.2: Minimum System Hardware and Environment Requirements

Component	Minimum Specification
Processor	Dual-core Intel / AMD processor
RAM	4 GB
Storage	5–10 GB free space
Operating System	Windows 10/11, Linux, or macOS
GPU	Not required
Internet	Required mainly for package installation and research

4.3 Experiments Conducted

This section describes the steps for data cleaning and analysis using two datasets viz. Dataset 1 and Dataset2.

4.3.1 Data Preprocessing

Dataset 1

- After data cleaning, six original variables and six engineered indexes: lifestyle indices (Mental_Wellness_Index, Social_Wellness_Index, Physical_Wellness_Index, Productivity_Index, Recovery_Wellbeing_Index, Stress_Burden_Index) were used for further data analysis. (See Table 4.3).
- The selected variables were standardized before applying PCA and K-Means.
- PCA was evaluated as a dimensionality-reduction method before final clustering to visualize structural pattern in data.
- K-Means clustering was evaluated for different values of k, followed by comparison of results obtained using the original and PCA-based feature spaces.

Dataset 2

- Identifier variable was removed (table : 4.3).
- Blood Pressure was separated into Systolic BP and Diastolic BP.
- Binary categorical variables were encoded numerically.
- Missing values and duplicate records were checked.
- Numerical variables were standardized before PCA and clustering.
- The scaler was fitted on the training data and subsequently applied to the testing data.

Table 4.3: Features used in two datasets for clustering.

Dataset 1	Dataset 2
Places_Visited, Donation, Bmi_Range, Sufficient_Income, Personal_Awards, Age, Mental_Wellness_Index, Social_Wellness_Index, Physical_Wellness_Index, Productivity_Index, Recovery_Wellbeing_Index, Stress_Burden_Index	BMI, Calories Intake, Hours of Sleep, Heart Rate, Alcohol Consumption per Week, Systolic BP, Diastolic BP

4.3.2 Engineered Feature Validation

The six engineered features (Mental_Wellness_Index, Social_Wellness_Index, Physical_Wellness_Index, Productivity_Index, Recovery_Wellbeing_Index, Stress_Burden_Index) were evaluated using range, variability, correlation and redundancy checks for Dataset 1. Table 4.4 shows the statistical description of these six newly engineered features, alongwith their correlation with selected target variable ‘Work-Life Balance’.

4.3.3 PCA for understanding structural patterns

In Dataset 1, the first principal component explained the largest proportion of variance. The **first three components together explained 51.47%** of the total standardized variance as vindicated in Figure 4.1. Although three components did not retain 90% of the variance, their usefulness was

subsequently evaluated through clustering performance rather than variance retention alone.

The results of the Principal Component Analysis indicate that the first three principal components explain a large part of the variance of the engineered features. Specifically, PC1, PC2 and PC3 accounted for 28.65, 12.41 and 10.41% of the variance, respectively. Overall PCA captures 51.47% of variance which is a lot considering that there are 6 engineered features in the data set (Figure 4.1). Thus, it can be concluded that all wellbeing indicators are informative enough to engineer the Work-Life Balance Score. They can be replaced by three new features for the purpose of dimensionality reduction.

In Dataset 2, the **first two components retained only 31.49% of the total variance**. The PCA projection showed substantial overlap between observations, with no clearly isolated groups or prominent outlier regions. This suggested that the health variables independently are insufficient to capture patterns in the data.

Table 4.4: Descriptive Statistics and Correlation with Work-Life Balance for Dataset 1

Engineered Feature	Descriptive Statistics		
	Mean	Std. Dev.	Pearson r
Productivity Index	0.487	0.217	0.685
Social Wellness Index	0.587	0.226	0.662
Mental Wellness Index	0.379	0.189	0.644
Recovery & Wellbeing Index	0.593	0.174	0.629
Physical Wellness Index	0.593	0.170	0.574
Stress Burden Index	0.410	0.193	-0.516

Principal Component	Explained Variance Ratio	Cumulative Variance
0	PC1	0.2865
1	PC2	0.4106
2	PC3	0.5147

Figure 4.1: Principal Component Analysis for Dataset 1

The descriptive statistics and correlation analysis show that most of the engineered wellbeing indicators tend to be positively correlated with the Work-Life Balance Score. Indeed, the highest positive correlation coefficient $r=0.685$ is exhibited by the Productivity Index. The Social Wellness Index

was $r=0.662$, and the Mental Wellness Index was $r=0.644$. But the Recovery & Wellbeing has a moderate positive relationship ($r=0.629$) and Physical Wellness has a positive relationship with $r=0.574$. At the same time, the Stress Burden Index is negatively correlated ($r=-.516$). The scatterplot also provides a graphical representation of the positive and negative trends of the Work-Life Balance Score against most of the indicators. There is a negative trend in particular, but positive trends outweigh the negative trend w.r.t target variable. Hence, it is sufficient to conclude that engineered features are positively related to an outcome variable (Table 4.4) .

4.3.4 Capturing structural patterns using clustering

We first details the clustering results and its profile on Dataset 1 and followed by second data Dataset 2.

Clustering results on Dataset 1

Table 4.5: Comparison of Clustering Schemes Across Original Features and Transformed Space using PCA on Dataset 1

Representation	Features	k	Silhouette $\uparrow$	DB $\downarrow$	CH $\uparrow$
Original	12	2	0.1606	2.0634	3509.94
PCA	3	2	0.3034	1.2439	8670.23
Original	12	5	0.1042	2.2314	1805.98
PCA	3	5	0.2434	1.1433	6053.45

Table 4.6: Clustering Metrics Comparison at Optimal $k = 2$ (Original vs. PCA)

Metric	Original (12 Features)	PCA (3 Components)
Silhouette Score $\uparrow$	0.1606	0.3034
Davies–Bouldin Index $\downarrow$	2.0634	1.2439
Calinski–Harabasz Score $\uparrow$	3509.94	8670.23

The PCA representation produced better clustering scheme than the original feature space for both tested values of k (see Table 4.5), hence PCA with 3 components, followed by K-Means clustering with $k=2$ was selected as the final clustering configuration, whose results are shown in Table 4.6.

Cluster Profiles

Main separating dimensions The largest centroid differences in Dataset 1 were observed for:

- Productivity Index
- Mental Wellness Index
- Recovery & Wellbeing Index
- Social Wellness Index
- Stress Burden Index

Table 4.7: Feature Means Across Clusters ($k = 2$) on Dataset 1

Feature	Cluster 0	Cluster 1
Places Visited	3.731	6.698
Donation	1.968	3.453
BMI Range	1.447	1.371
Sufficient Income	1.600	1.859
Personal Awards	4.302	7.096
Age Category	1.435	1.778
Mental Wellness Index	0.277	0.479
Social Wellness Index	0.474	0.699
Physical Wellness Index	0.520	0.665
Productivity Index	0.366	0.607
Recovery & Wellbeing Index	0.501	0.684
Stress Burden Index	0.502	0.318

Table 4.7 presents the comparison of the mean values of features between the two clusters. It can be observed that Cluster 1 has a higher share of most positive well being-related indicators. It includes the number of countries visited, donations, sufficient income, and personal awards, higher values of age category, and all the well being indices except the Stress Burden Index. Specifically, the Mental, Social, Physical, Productivity, and Recovery and Well being indices are better to capture structure in the data.

On the contrary, the Stress Burden Index is higher (0.502 vs. 0.318) in Cluster 0, which may indicate that the participants in this cluster experience higher levels of stress despite a similar range of BMI. Thus, it can be concluded that the sample clusters are different from each other as one cluster (Cluster 1) has higher wellbeing indicators while the other (Cluster 0) has a

higher Stress Burden index. However, this relationship should be interpreted as an association, not causation.

Clustering results on Dataset 2

Initially, k-means clustering was performed with $k=4$ on Dataset 2 whose details are given in Table 4.8.

Table 4.8: Cluster Distribution and Sample Breakdown ($N = 800$) on Dataset 2

Cluster	Observations (n)	Percentage (%)	Cluster Compactness
Cluster 0	186	23.25	5.6657
Cluster 1	199	24.88	6.7666
Cluster 2	207	25.88	6.1329
Cluster 3	208	26.00	6.1519
Total	800	100.00	

It is vindicated from Table 4.8 that the generated four clusters were relatively balanced in terms of their size. The relatively similar diameters indicate that no cluster had an extreme difference in internal spread. However, Cluster 0 had the smallest diameter, while Cluster 1 had the largest.

Table 4.9: Evaluation of Clustering Quality on Dataset 2

Metric	Result
Number of Clusters (k)	4.0000
Overall SSE / Inertia	4235.2700
Silhouette Score	0.1077
Diabetes Cluster Purity	0.8450
Heart Disease Cluster Purity	0.9087

We also applied DBScan clustering on Dataset 2 with $\epsilon = 2$ and minimum samples as 14. But, it generated a single cluster with only three noise observations. DBSCAN did not provide a useful multi-cluster representation of the dataset under the tested configuration. Hence, K-Means clustering with $k = 4$ was selected as the final clustering model for further analysis. Standardized feature centroids across 4 clusters are shown in Table 4.10

Figure 4.2 shows the mean of each feature cluster-wise. Based on the Table 4.10 and Figure 4.2, following observations are made for Dataset 2:

Table 4.10: Standardized Feature Centroids Across Clusters ($k = 4$)

Feature (Z -score)	Cluster 0	Cluster 1	Cluster 2	Cluster 3
BMI	−0.919	0.769	−0.364	0.448
Calories Intake	−0.116	0.186	−0.111	0.037
Hours of Sleep	0.767	0.811	−0.779	−0.687
Heart Rate	−0.211	0.634	0.209	−0.626
Alcohol Consumption	0.419	0.024	0.006	−0.404
Systolic BP	−0.483	0.224	0.790	−0.569
Diastolic BP	0.143	0.021	0.248	−0.395

- **Cluster 0** consists individuals who are lean, relatively well-rested with lower blood pressure and comparatively higher alcohol consumption.
- **Cluster 1** consists of individuals having Higher-BMI with adequate sleep but elevated heart rate and somewhat higher blood pressure.
- **Cluster 2** consists of individuals who are characterized mainly by insufficient sleep and elevated blood pressure. This is the cluster with the strongest blood-pressure-related profile.
- **Cluster 3** consists of individuals with moderately higher BMI, reduced sleep, lower heart rate and lower blood-pressure measurements.

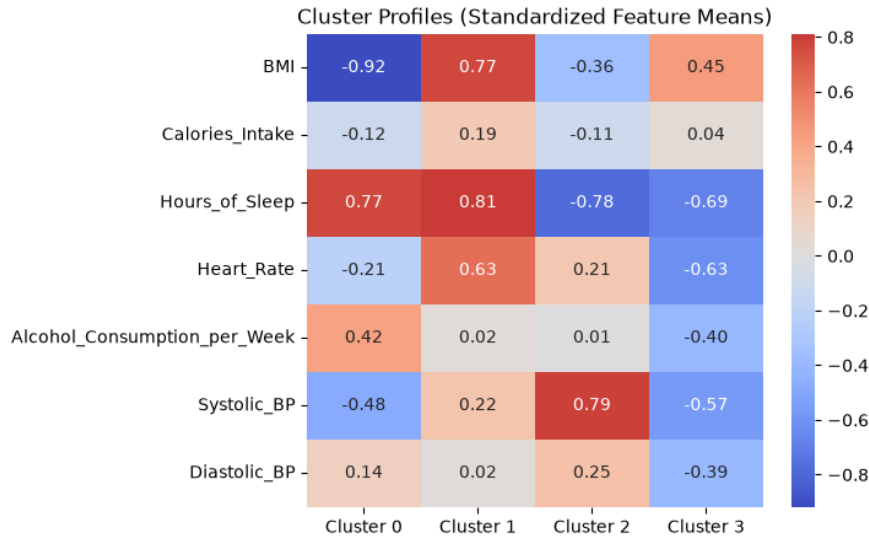


Figure 4.2: Cluster Profile of Dataset 2

4.3.5 Key Findings

This section explains the key findings for Dataset 1 and Dataset 2. Following are the key findings and important factors w.r.t. Dataset 1.

i. Engineered Features:

- All six engineered indices were successfully transformed to a common 0–1 scale.
- All six showed meaningful variability.
- Their correlations with Work-Life Balance followed the expected directions (fig : 4.4).
- Productivity showed the strongest positive Pearson correlation (0.685) (table 4.4).
- Stress Burden showed a negative correlation (-0.516) (fig : 4.4).

ii. Principal Component Analysis (PCA):

- PC1 explained 28.65% of the variance.
- The first three components explained 51.47% of the cumulative variance (fig : 4.1).
- PCA reduced the clustering representation from 12 variables to 3 principal components (table : 4.5).
- PCA significantly improved clustering metrics compared with the original feature space.

iii. Clustering Evaluation:

- The final model was PCA + K -Means with $k = 2$ (fig : 4.3).
- Silhouette Score: 0.3034
- Davies–Bouldin Index: 1.2439
- Calinski–Harabasz Index: 8670.23
- The original and PCA-based $k = 2$ assignments showed an Adjusted Rand Index (ARI) of 0.9910.

iv. Cluster Structure and Profile:

- Cluster 0 contained 49.60% of observations.
- Cluster 1 contained 50.40% of observations.
- Cluster 1 exhibited higher values across the wellbeing, productivity, and recovery indices.

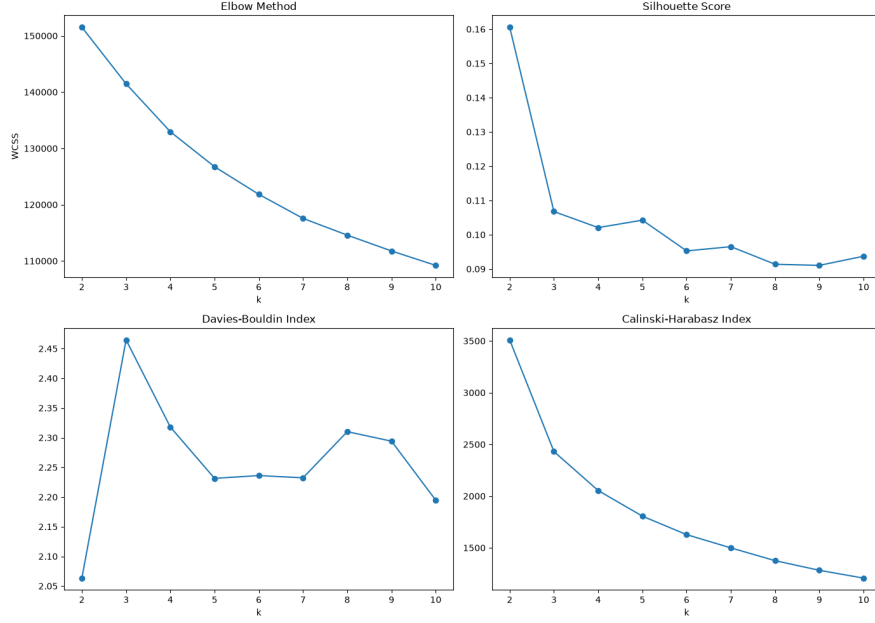


Figure 4.3: Clustering Evaluation using four indexes on Dataset 1

- Cluster 0 exhibited a higher Stress Burden Index.

For Dataset 1, The largest separation was observed for features in the order of **Productivity** → **Mental Wellness** → **Recovery & Wellbeing** → **Social Wellness** → **Stress Burden**.

We also detail below the major findings for Dataset 2.

i. Principal Component Analysis (PCA):

- PCA was initially applied to 12 standardized continuous variables.
- PC1 explained 10.34% of the variance.
- PC1 + PC2 explained 20.02% of the cumulative variance (fig :4.4).
- Ten components retained approximately 85.87% of the total variance.
- PCA applied to the seven clustering variables retained 31.49% in the first two components.
- A reduced five-feature PCA representation increased PC1 + PC2 variance to 43.26% (see Figure 4.4).

ii. Clustering Configuration and Evaluation:

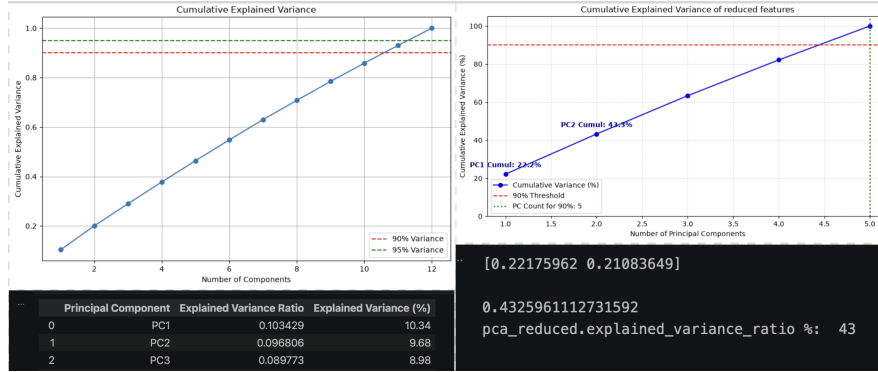


Figure 4.4: Principal Component Analysis for Dataset 2

- The final clustering feature set contained 7 standardized health variables.
- K -Means was evaluated across $K = 2$ –10.
- The final selected solution was $K = 4$.
- Overall SSE / Inertia: 4235.27.
- Silhouette Score: 0.1077.
- The resulting clusters were approximately balanced in sample size.

iii. Cluster Profiles:

- The four clusters were primarily differentiated by:
 - Body Mass Index (BMI)
 - Hours of Sleep
 - Blood Pressure (Systolic and Diastolic)
 - Heart Rate
 - Alcohol Consumption

iv. Disease Alignment and Clinical Validation:

- Diabetes cluster purity: 84.50%.
- Heart Disease cluster purity: 90.87%.
- *Critical Note:* These purity values must not be conflated with classification/prediction accuracy; underlying class imbalance contributes substantially to the observed high purity.

The clustering on Dataset 2 did not produce sharply separated geometric groups, but the resulting four clusters demonstrated substantial alignment with health-risk outcomes, with heart disease showing particularly strong cluster-level purity, as mentioned above.

4.4 Result Analysis

The analysis of first dataset (Dataset 1) suggests that different lifestyle and well being factors were associated with work life balance. The six engineered wellness features showed meaningful relationships with work-life balance Score, suggesting that combining related variables can provide a better understanding than analyzing individual variables separately. Among these features, productivity showed the strongest positive correlation with Work-Life Balance [8]. In PCA analysis, it is observed that the first three components captured about half of the information in the dataset, which isn't enough to explain 90% of the data, but they were still tested to see weather they could effectively update the data into meaning full clusters. The analysis also suggested an overlap between the stress burden and recovery & well-being which suggest that these two features may contain similar information. Some lifestyle variables(e.g- daily shouting, lost vacation etc) were unevenly distributed.

The second dataset Dataset 2 [9] used also revealed similar information. The application of PCA did not prove to be useful, as the information was uniformly distributed across several factors rather than being concentrated in only a few dimensions. This also highlights a limitation of synthetic data generator. K-Means analysis on this data resulted in four clusters, but the distinction between these groups was not very strong (Figure 4.5).

Close inspection of clusters w.r.t. diabetes and heart disease did not provide a strong basis for interpreting the cluster as distinct health groups. Overall, the results suggests that the health-related characteristics are influenced by a combination of lifestyle and physiological factors (like- blood pressure, cholesterol etc.) rather than forming simple and clearly defined groups.

When both datasets Dataset1 and Dataset 2 are studied together, a common pattern is observed. The multivariate analysis in both datasets showed that the overall structure was complex and may not be captured using standard EDA techniques. Therefore, the findings support the idea that lifestyle, well-being and health are interconnected through multiple factors. This highlights the importance of considering multiple lifestyle and physiological factors together when analyzing health-related patterns and require deep



Figure 4.5: Cluster identification in Dataset 2

learning techniques to capture complex patterns.

Chapter 5

Conclusion and Future Scope

This chapter presents the main findings of the current study along with limitations and its future scope.

5.1 Conclusion

The study analyzed two datasets related to two lifestyle factors viz. well-being health and physical activity [8, 9]. Both datasets were examined to understand the relationship between different factors and to identify broader patterns within the data. It is observed that simple EDA is insufficient to reveal meaningful information. Therefore, the findings support that lifestyle, well-being and health are interconnected through multiple factors, which require careful and complex analysis considering multiple lifestyle and physiological factors together.

5.2 Limitations

The following are the limitations of the work carried out.

- **Cross sectional nature of dataset:** We can identify association and lifestyle patterns, but were unable to establish temporal or casual relationships because of the non-availability of temporal data.
- **Self reported questionnaire dataset:** This is one of the biggest limitations of our project: most of the variables are ratings collected through Google form or questionnaires which means the measurement depends on participant perception and reporting.

- **Engineered Features are constructed according to understanding:** The constructed features were not validated through medical or psychological instruments, whereas we have constructed them by combining features according to our conceptual framework.
- **Dataset 2 was a synthetic data set:** As Dataset 2 is a synthetic data set, hence it has a high risk of not capturing real patterns in health related factors.
- **No external clinical validation:** As data set is obtained from questionnaires and features are engineered but the external clinical validations like BP, diabetes variables are present in data set one [8] which lack in establishing the relationship between lifestyle variable and clinical variables.

5.3 Future Scope

Although the present investigation contributes meaningfully to an understanding of connections between health indicators and aspects of lifestyle and wellness, there are multiple promising avenues for future work, including the following:

- **Diverse Datasets:** In this study we used the available datasets, but in future we can collect larger and diverse datasets so that the results can be compared across different age group of people, different occupations and geographical backgrounds.
- **Collect data for longer intervals:** It is also required to collect data from the same individuals for the longer intervals, Which can help in understanding the impact of changes in lifestyle and physiological factors on individuals well-being and health outcomes over time.
- **Explore Unsupervised Deep Learning Methods:** Both K-means and DBSCAN could not reveal prominent patterns in data. Hence, need is to leverage unsupervised deep learning approaches to reveal complex patterns.
- **Validate the findings with real-world data:** Future analysis could also demographic based real data to reveal complex patterns in health factors and life style.

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